

Quiz 6 Class 2 Solutions

Computer Security Fundamentals Spring 2026

Instructions: Answer the questions, **closed-book, closed-notes**. Time: 10 minutes

1. (**F**) Consequentialist ethical framework judges an action based strictly on its adherence to duties or rules, independent of the outcomes. (True/False, 2 pts)

2. (**C**) An engineer wants to discover malware running in a laptop using side channels. Which option below does **NOT** belong to side channel information? (2 pts)

- A. The CPU core temperature rises sharply.
- B. The battery of the laptop drains much faster.
- C. The OS shows privilege warnings repeatedly.
- D. The hard drive access time becomes much longer.

3. (**B**) Which option best describes the differences between symbolic execution and dynamic testing? (2 pts)

- A. Symbolic execution cannot handle loops in the source code, but dynamic testing can.
- B. Symbolic execution can represent many dynamic testing runs in one execution.
- C. Symbolic execution does not suffer from path explosion, but dynamic testing does.
- D. Symbolic execution is always complete, but dynamic testing can be incomplete.

4. (Short Answer, 4 pts) Perform symbolic execution for the source code below to determine the exact path condition required to execute the abort(); statement on line 7. Provide concrete values of x and y that would satisfy the condition, one each.

```

1  int x = read_input(); // Symbolic value alpha
2  int y = read_input(); // Symbolic value beta
3  int z = x * 3;
4  if (z > 12) {
5      y = z + y;
6      if (y <= 5) {
7          abort(); // Target Line
8      }
9  }
```

Solution:

Line 4 $z < 12 \rightarrow x * 3 > 12 \rightarrow x > 4$

Line 6 $y \leq 5 \rightarrow z + y \leq 5 \rightarrow x * 3 + y \leq 5$

Final path condition for line 7: $x > 4 \ \&\& \ 3x + y \leq 5$

One possible pair for x and y: $x == 5, y == -10$